

Managerial Accounting & Costing Systems

What students should be able to do after this session

- Classify costs for different managerial purposes: cost objects, manufacturing, reporting, behavior, and decision making.
- Distinguish direct materials, direct labor, manufacturing overhead, selling cost, and administrative cost.
- Explain product costs, period costs, variable costs, fixed costs, mixed costs, and cost drivers.
- Apply job-order costing using job cost sheets, source documents, and predetermined overhead rates.
- Compare job-order costing with process costing and compute equivalent units and costs per equivalent unit.

Teaching purpose

This deck combines three chapters into one storyline: first understand cost language, then assign costs to unique jobs, then assign costs to continuous processes.

Class rhythm

Definition → example → calculation → flow of costs → managerial interpretation.

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Integrated roadmap for the class

1

Cost concepts

What cost are we talking about, and for what purpose?

2

Cost behavior

How does cost change when activity changes?

3

Job-order costing

How do we cost customized jobs or orders?

4

Process costing

How do we cost homogeneous units produced continuously?

Core idea: cost systems translate production activity into product cost information for reporting and decisions.

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01

Cost concepts and cost behavior

Before assigning costs, students must know what type of cost is being assigned and why.

Different cost classifications serve different purposes

Purpose	Classification	Key question
Assigning costs to objects	Direct cost vs indirect cost	Can we trace the cost conveniently to a product, customer, job, or department?
Manufacturing companies	Direct materials, direct labor, manufacturing overhead	What costs are incurred to make the product?
Financial statements	Product costs vs period costs	Is the cost inventoried first or expensed immediately?
Cost behavior	Variable, fixed, mixed	How does total cost respond to activity?
Decision making	Differential, opportunity, sunk	Which costs should affect a choice among alternatives?

Direct cost vs indirect cost

Cost object

Anything for which cost data are desired: product, customer, job, department, project, or organizational subunit.

Direct cost

A cost that can be easily and conveniently traced to a specified cost object.

Indirect cost

A cost that cannot be easily and conveniently traced to one specified cost object.

Question **Is factory electricity a direct cost or indirect cost?**

Manufacturing costs: the three product-cost building blocks

Direct materials

Raw materials that become an integral part of the finished product and can be conveniently traced to it.

Direct labor

Labor costs that can be easily traced to individual units of product; often called “touch labor.”

Manufacturing overhead

All manufacturing costs other than direct materials and direct labor: indirect materials, indirect labor, utilities, depreciation, maintenance.

Total manufacturing cost = Direct materials + Direct labor + Manufacturing overhead

Prime cost and conversion cost

Prime cost = Direct materials + Direct labor

Conversion cost = Direct labor + Manufacturing overhead

Why prime cost matters

Highlights the direct inputs that can be traced to the product.

Why conversion cost matters

Captures the cost to convert materials into finished goods. It becomes very important in process costing.

Prime = direct inputs. Conversion = transformation effort.

Nonmanufacturing costs: selling and administrative

Selling costs

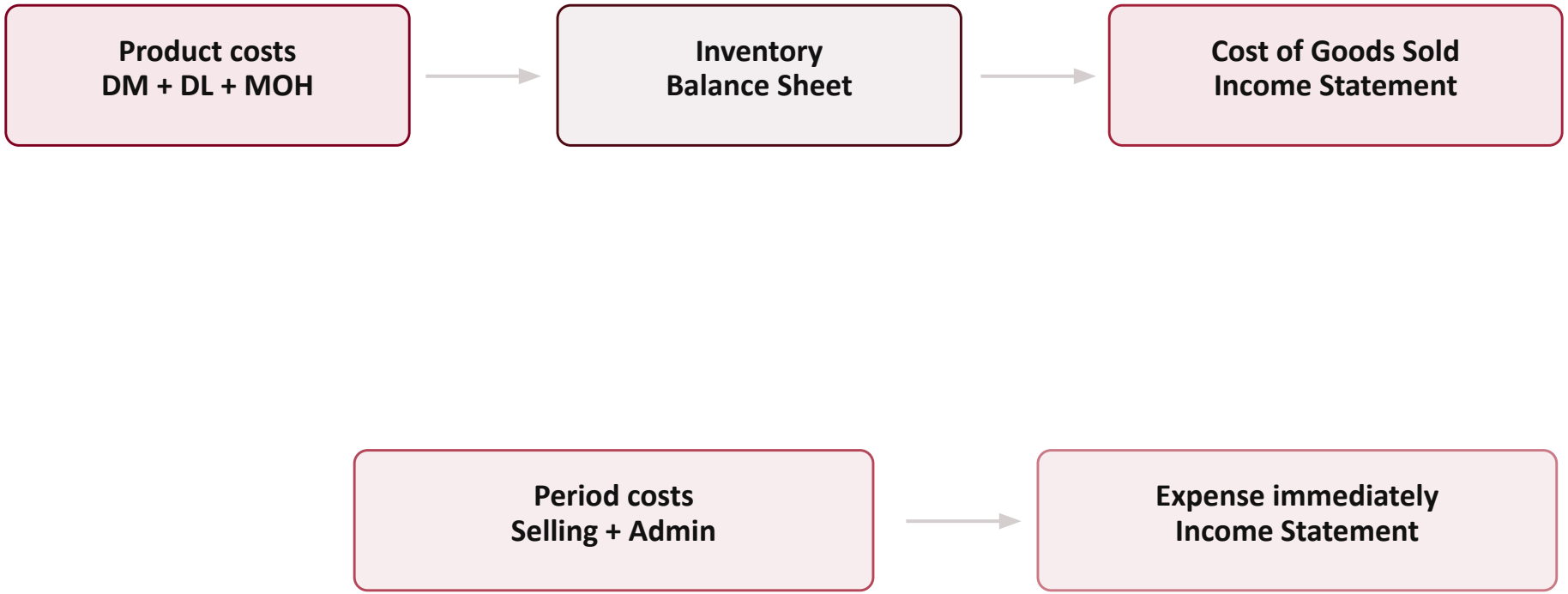
All costs necessary to secure customer orders and get the finished product to customers: advertising, sales commissions, shipping, sales travel, finished goods warehouse costs.

Administrative costs

Executive, organizational, and clerical costs associated with general management: accounting, secretarial, public relations, executive compensation.

Manufacturing costs become product costs; selling and administrative costs are period costs.

Product costs vs period costs



Product costs wait in inventory until sale; period costs are expensed in the period incurred.

Variable, fixed, and mixed costs

Type	Total cost behavior	Per-unit behavior	Example
Variable cost	Changes in direct proportion to activity	Constant per unit	Cost per text message; direct material per unit
Fixed cost	Constant within the relevant range	Decreases per unit as activity rises	Monthly rent; depreciation; factory supervisor salary
Mixed cost	Contains both fixed and variable components	Depends on cost equation	Utility bill: fixed charge + cost per kWh

Mixed cost equation: Y = a + bX

Where

Y = total mixed cost; a = fixed cost; b = variable cost per activity unit; X = level of activity.

Activity base or cost driver

Definition

A measure of what causes the incurrence of variable costs. As the activity base increases, total variable cost increases proportionally.

Common cost drivers

Units produced or sold, machine hours, labor hours, miles driven, patient-days, kilowatt hours, orders processed.

For a hospital, is patient-days a better cost driver than number of doctors? Why?

Estimating mixed costs: three techniques

Method	How it works	Strength / caution
Scattergraph	Plot activity and cost; fit a visual line to estimate fixed and variable components.	Useful for checking linearity, but subjective.
High-low	Use highest and lowest activity points to estimate variable cost per unit and fixed cost.	Fast, but depends on only two data points.
Least-squares regression	Uses all data points to fit a line that minimizes squared errors.	Stronger statistically; check R^2 and reasonableness.

High-low variable cost per unit = Change in cost ÷ Change in activity

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Decision-making cost classifications

Differential cost / revenue

Difference in costs or revenues between alternatives. Relevant because it changes across choices.

Opportunity cost

Potential benefit given up when one alternative is selected over another. Often not recorded in accounting systems.

Sunk cost

Cost already incurred that cannot be changed now or in the future. Ignore for decisions.

Relevant decision rule: focus on future costs and benefits that differ among alternatives.

02

Job-order costing

Assign costs to unique jobs using source documents, job cost sheets, and predetermined overhead rates.



When job-order costing is appropriate

Many different products or jobs are produced each period.
Products or services are manufactured to order or customized for customers.
Each order is unique enough that costs must be traced or allocated to each job separately.
Common examples: aircraft manufacturing, construction projects, consulting projects, legal cases, movie production.

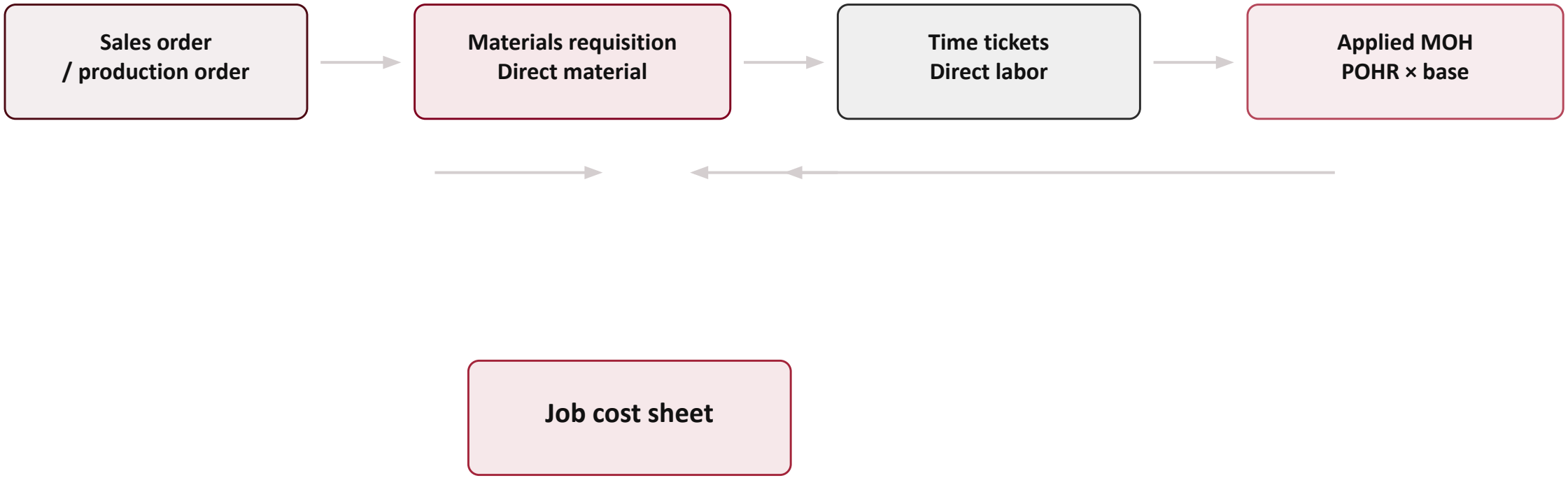
Cost object

The individual job becomes the main cost object.

Main document

The job cost sheet accumulates direct materials, direct labor, and applied overhead for a specific job.

Job-order costing flow



Total job cost = Direct materials + Direct labor + Applied manufacturing overhead

Job cost sheet: the subsidiary ledger

Section	What is recorded	Source document
Direct materials	Materials issued to a specific job	Materials requisition form
Direct labor	Hours and wage cost charged to a job	Employee time ticket
Manufacturing overhead	Overhead applied using predetermined overhead rate	POHR × actual allocation base
Cost summary	Total cost and unit product cost	Job cost sheet summary

The job cost sheet supports the Work in Process account by showing costs for each individual job.

Why overhead is applied, not traced

Reason 1

Many overhead costs cannot be traced conveniently to a specific job.

Reason 2

Overhead includes many different items: indirect materials, indirect labor, production manager salary, utilities, depreciation, maintenance.

Reason 3

Actual overhead is not known until period-end and may fluctuate seasonally.

Predetermined overhead rate (POHR) = Estimated total manufacturing overhead ÷ Estimated total allocation base

Applied overhead = POHR × Actual allocation base used by the job

POHR computation example

Given

Estimated fixed MOH = $200,000
Variable MOH = $2.75 per DLH
Estimated allocation base = 160,000 DLH

Step 1: estimate total MOH

Y = a + bX
Y = $200,000 + ($2.75 × 160,000)
Y = $640,000

Step 2: compute POHR

$640,000 ÷ 160,000 DLH
= $4.00 per direct labor-hour

If Job A-143 used 8 DLH, applied overhead = 8 × $4 = $32

Unit product cost: useful, but not incremental cost

Unit product cost = Total job cost ÷ Units completed

Example

If total job cost is $236 for 2 crates, average unit product cost is $118.

Important caution

Average unit cost is not always the cost of producing one additional unit. Fixed overhead may not change for one more unit.

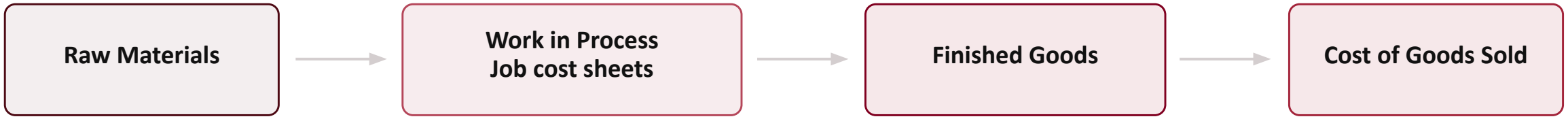
Why can average unit product cost mislead pricing or special-order decisions?

Journal entry logic: how costs flow

Transaction	Debit	Credit	Meaning
Purchase raw materials	Raw Materials	Accounts Payable / Cash	Materials enter inventory
Use direct materials	Work in Process	Raw Materials	Direct materials charged to job
Use indirect materials	Manufacturing Overhead	Raw Materials	Indirect materials collected in MOH
Direct labor incurred	Work in Process	Wages Payable	Labor traced to job
Indirect labor incurred	Manufacturing Overhead	Wages Payable	Labor collected in MOH
Apply overhead	Work in Process	Manufacturing Overhead	Overhead assigned to jobs

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Inventory flow: WIP → Finished Goods → COGS



COGM

Cost of goods manufactured is the cost transferred from WIP to Finished Goods when jobs are completed.

COGS

Cost of goods sold is the cost transferred from Finished Goods to expense when completed jobs are sold.

COGM and COGS formulas

Measure	Formula
Raw materials used	Beginning raw materials + Purchases – Ending raw materials
Direct materials	Raw materials used – Indirect materials included in MOH
Total manufacturing costs	Direct materials + Direct labor + Manufacturing overhead applied to WIP
Cost of goods manufactured	Total manufacturing costs + Beginning WIP – Ending WIP
Unadjusted COGS	Beginning finished goods + COGM – Ending finished goods
Adjusted COGS	Unadjusted COGS ± underapplied/overapplied overhead

03

Process costing

Assign costs to homogeneous units produced continuously through departments.

When process costing is appropriate

A single product or homogeneous units are produced continuously for a long period of time.
Production occurs through one or more processing departments.
Costs are accumulated by department, not by individual job.
Unit cost is computed by department using equivalent units of production.

Examples

Beverages, chemicals, cement, food processing, paper, oil refining, and other continuous production settings.

Main document

Department production report: summarizes units, equivalent units, costs per equivalent unit, assigned costs, and reconciliation.

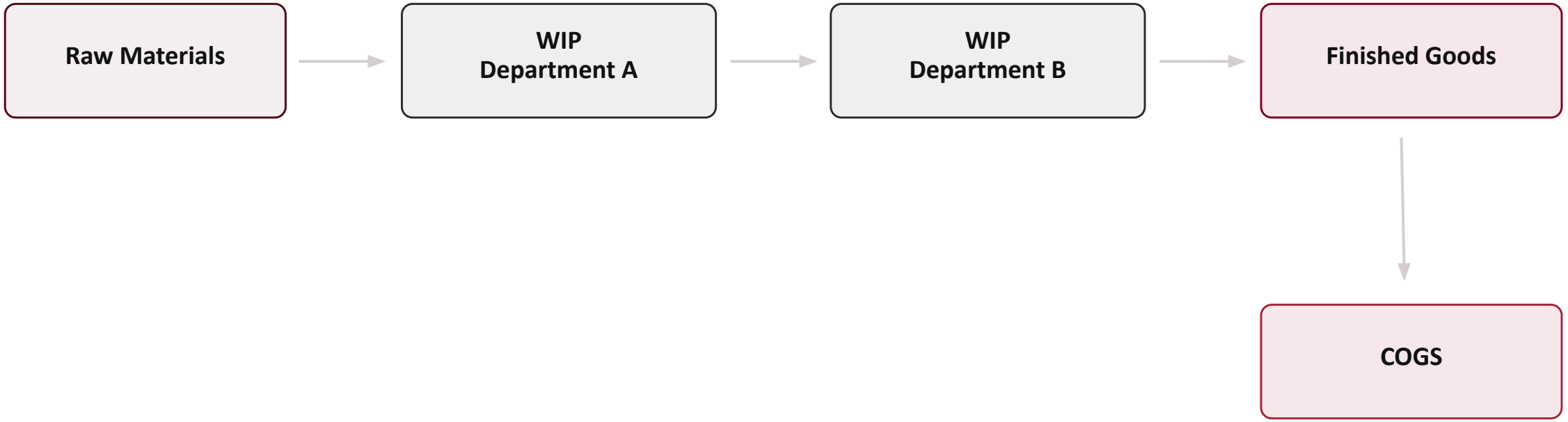
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Job-order vs process costing

Dimension	Job-order costing	Process costing
Production setting	Customized jobs; many different products	Homogeneous units; continuous production
Cost accumulation	By individual job	By processing department
Main document	Job cost sheet	Department production report
Unit cost	Computed by job	Computed by department
WIP account	Often one WIP with job sheets as subsidiary records	Separate WIP account for each department

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Processing departments and cost flow



Key term

Transferred-in costs are costs moved from one processing department to the next.

Equivalent units: the core idea

Two half-completed units = one equivalent complete unit

Example

10,000 units that are 70% complete are equivalent to 7,000 complete units.

Why it matters

Process costing must assign costs between units completed and units still in ending work in process.

Equivalent units = Partially completed units × Percentage completion

Equivalent units of production

Production flow equation

Beginning WIP units + Units started or transferred in = Ending WIP units + Units completed and transferred out

Cost flow equation

Cost of beginning WIP + Costs added during the period = Cost of ending WIP + Cost of units transferred out

Equivalent units of production = Units transferred out + Equivalent units in ending WIP

A separate equivalent-unit calculation is usually made for each cost category, especially materials and conversion costs.

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Materials vs conversion costs

Direct materials

Materials may be added at the beginning, at the end, or gradually throughout a department's process. Completion percentage for materials can differ from conversion.

Conversion costs

Direct labor and manufacturing overhead may be combined because direct labor can be small relative to other product costs in automated process environments.

Conversion costs = Direct labor + Manufacturing overhead

Cost per equivalent unit

Cost per equivalent unit = (Beginning WIP cost + Costs added during period) ÷ Equivalent units of production

Cost category	Cost pool	Divide by
Materials	Beginning material cost + material cost added	Material equivalent units
Conversion	Beginning conversion cost + conversion cost added	Conversion equivalent units
Department total	Material CPEU + Conversion CPEU	Total cost per equivalent complete unit

Apply costs and reconcile

1. Ending WIP

Equivalent units in ending WIP × cost per equivalent unit, computed separately for materials and conversion.

2. Units transferred out

Units completed and transferred × cost per equivalent unit. Units transferred out are assumed 100% complete for that department.

3. Reconciliation

Total costs accounted for must equal total costs to be accounted for.

Beginning WIP cost + Costs added = Ending WIP cost + Cost transferred out

Operation costing: the hybrid

Job-order extreme

Unique jobs, customized products, costs tracked by job.

Operation costing

Hybrid system used when products have some common processing and some individual customization.

Process extreme

Homogeneous products, continuous flow, costs tracked by department.

Teaching point

Real-world costing systems often sit between textbook categories.

04

Teaching cases and practice

Use these closing slides to move from definition to application and diagnosis.

Mini-case 1: classify the costs

Cost item	Student task
Wood used in custom furniture order	Direct or indirect? Product or period?
Factory supervisor salary	Manufacturing or nonmanufacturing?
Sales commission	Product or period?
Head office accounting staff	Manufacturing or nonmanufacturing?
Depreciation on factory equipment	Direct or indirect?

Mini-case 2: choose the costing system

Scenario A

An advertising agency designs a campaign specifically for one client.

Scenario B

A beverage manufacturer produces the same bottled tea continuously.

Scenario C

A manufacturer uses a common production process, then customizes final packaging for each customer.

Mini-case 2: choose the costing system (answered)

Scenario A

An advertising agency designs a campaign specifically for one client.

Scenario B

A beverage manufacturer produces the same bottled tea continuously.

Scenario C

A manufacturer uses a common production process, then customizes final packaging for each customer.

Discuss! Which costing system relevant for those scenarios and why ?

Mini-case 3: overhead application

Given

Estimated total manufacturing overhead = $480,000
Estimated machine hours = 120,000
Job X used 35 machine hours

Student calculation

1. Compute POHR
2. Apply overhead to Job X
3. Explain why actual overhead is not used immediately

Mini-case 3: overhead application (answered)

Given

Estimated total manufacturing overhead = $480,000
Estimated machine hours = 120,000
Job X used 35 machine hours

Student calculation

1. Compute POHR
2. Apply overhead to Job X
3. Explain why actual overhead is not used immediately

Answer

POHR = $480,000 ÷ 120,000 = $4/MH
Applied overhead = 35 × $4 = $140

Mini-case 4: equivalent units

Given

Ending WIP = 800 units
Materials = 60% complete
Conversion = 25% complete

Student task

Compute equivalent units for materials and conversion.

Mini-case 4: equivalent units (Answered)

Given

Ending WIP = 800 units
Materials = 60% complete
Conversion = 25% complete

Student task

Compute equivalent units for materials and conversion.

Answer

Materials EU = 800 × 60% = 480
Conversion EU = 800 × 25% = 200

Equivalent units must be computed separately when completion percentages differ by cost category.

What students should remember

Cost classification depends on purpose; the same cost may be classified differently in different contexts.
Product costs become inventory first, then cost of goods sold; period costs are expensed immediately.
Job-order costing traces direct costs to jobs and applies overhead using a predetermined rate.
Process costing accumulates costs by department and uses equivalent units to assign costs between completed units and ending WIP.
Cost systems are not just bookkeeping tools; they support pricing, budgeting, performance evaluation, and operational decisions.

Next class bridge

Use the product cost information from Week 4 to support planning, control, and managerial decision-making.